

SI 649 011 WN 2026



Edit

SI649/EECS548 Winter 2026 (Section 011/012/111/112) Course Syllabus and FAQ

1. Who, What, Where, When

Course: SI649 / EECS548, Information Visualization, Winter 2026 (Section 011/012/111/112)

Instructor: Dallas Card (dalc@umich.edu); Office Hours: 3-4pm on Thursdays in LCSIB 3549

GSIs:

- Yelnaz Rysbek (yrysbek@umich.edu); Office Hours: 1:00-3:30pm on Wednesdays (in person or online), scheduled here: <https://calendly.com/yrysbek-umich/si-649-office-hours> ↗ (<https://calendly.com/yrysbek-umich/si-649-office-hours>)

Lecture schedule:

- Lecture: Thursday 4:30-6:00pm LCSIB 2290 (starting Jan 8)

Labs / discussion schedule:

- Lab: Tuesdays 4:30-6:00pm LCSIB 2290 (starting Jan 13)

Class format: This will be an in-person class, and it is expected that you will regularly attend lectures and labs. Reasonable accommodations will be made when people are unable to attend due to illness or travel.

Contacting us: We will be using Piazza for class discussions, with a link available on the Canvas page for this course. Please be sure to sign up for this and participate in the discussions. As a backup option, you can email the instructors, but Piazza will likely get a faster response.

2. Summary

This class is a 14-week course on Information Visualization (plus one week for winter break). Most weeks there will be required readings, on which you will be expected to read and comment using Perusall, due the day before each lecture.

Lectures will be a mix of course material, discussion, and in-class group work. Lab / discussion sessions will be a mix of tutorials, design jams, presentations, and sharing peer feedback. (Some weeks there will be no lab to reserve time for working on projects).

Most labs will have a corresponding assignment, due one week later, and time will be made available in the lab to get started on these. These assignments will be a mixture of creating visualizations (using the tools covered in the labs), research/writing assignments, and reflections. In addition, there will be a solo project and a group project. Project deadlines will mostly be on Sunday nights (see schedule for details).

Throughout the term, we will touch on a variety of visualization tools, including matplotlib, Altair, p5.js, and others. For projects, you are welcome to use other tools if you prefer.

3. Learning Objectives

The primary objectives for this course are to learn to understand, interpret, critique, design, and create information visualizations -- that is, ways of visualizing abstract data so as to make sense of, communicate, and support decision making about real world information. These learning objectives will be developed through lectures, labs, readings, assignments, projects, and in-class discussions and activities. Students will learn the theory and principles underlying information visualization, how to read the literature and critique the work of others, and how to use a variety of tools, including programmatic tools, to create their own visualizations.

4. Grading

The weight for each grading component is given in the table below:

Component	Weight
Social Annotations of Weekly Readings	10%
Weekly Assignments	15%
Static visualization project	5%
Interactive redesign project	10%
Narrative visualization project	40%
Final exam	10%
Participation	10%

5. Late Policy

Please try your best to submit all work on time. For social annotations of the readings, no late submissions will be accepted (because we will need time to review them before the lectures), but you are free to skip two of these during the course (the best N-2 will be used if you complete them all). For assignments and project deadlines, a late turn-in within 24 hours will cost you one letter grade or equivalent (so the best you'll be able to get is a "B"). A later turn in between 24 and 48 hours will cost you 2 letter grades. We may also offer a make up assignment for those who miss one.

6. Coursework

There are five components of this course in which you will be evaluated.

i. Class participation

Participating in this class means attending lectures and labs, paying attention, asking and answering

questions, actively participating in in-class discussions and activities, giving feedback to others, and helping out your fellow students. This class will involve a lot of informal group work, discussions, and presentations. I will call on people randomly in some cases, but it is expected that you will participate, without having to be called upon. You will get some credit for simply attending class regularly, but if you *only* attend without actively participating, your participation grade will suffer.

ii. Weekly readings

Most weeks will have required readings (as well as optional readings), available through the Canvas site. For the required readings, you will be expected to provide social annotations using Perusall. This includes asking questions about the material, commenting on other people's questions, providing links to external resources, pushing back on what is in the readings, or adding your own perspective. In past years, this has led to lots of positive engagement from students.

Please do your best to engage seriously with these readings. A few deep comments are better than many superficial comments. Comments throughout the readings are better than only commenting on one section. 2-4 comments that show you have thought about the material will typically be sufficient for you to get 4 out of 5 points. Full credit will be given to those who go above and beyond. Reading annotations will be due on Wednesday nights.

iii. Weekly Assignments

Most weeks there will be an assignment to complete. Some of these will involve creating information visualizations using the tools that will be covered in the lab. Others will involve reflection, peer review, or some amount of research or writing. Additional details will be provided in advance of each assignment. For all of these assignments, you should feel free to discuss the problems and brainstorm with your colleagues, but you should always write up your own solution, code, and/or reflection. Especially for code, you should make sure you understand not just how to produce the desired result, but why it works (and there is always more than one way to do something). Please add comments in the code if you are unsure about something, and ask for help as necessary. Weekly assignments will be due one week after they are assigned, before the start of the lab on Tuesday.

iv. Projects

The first project in the course will be a solo project, which will be made up of two stages. In the first stage, you will work with a given dataset to produce static visualizations. In the second stage, you will workshop your static visualizations with others, and create interactive redesigns of those.

The final project in this course will be a group project, for which you will be allowed to choose your own teams. (We will provide assistance if you are unable to find a team on your own). This project will be a narrative visualization that you will produce on a topic of your choice, along with accompanying the accompanying text, based on your own research. Project deadlines will typically be on Sunday nights. More details will be provided on these projects as the course proceeds.

v. Final Exam

This course will also have an in-person, on-paper, final exam, that will test your knowledge of concepts covered throughout the course. To avoid conflicts due to travel during the exam period, this exam will be held

during the last lecture period of the class.

7. Schedule

A (tentative) course schedule is given in the table below. Specific timing and topics are subject to revision.

Week	Lab Date (Tuesday)	Lab Topic	Lecture Date (Thursday)	Lecture Topic	Project Deadlines (due date)
1			Jan 8	Introduction	
2	Jan 13	First critique	Jan 15	Data and Image Models	
3	Jan 20	matplotlib	Jan 22	Design and Aesthetics	
4	Jan 27	Design workshop	Jan 29	Perception	Static viz project ideas (Feb 1st)
5	Feb 3	Altair basics	Feb 5	Color and Evaluation	
6	Feb 10	Altair transformations	Feb 12	Communication and ethics	Static viz project (Feb 15th)
7	Feb 17	Project share and critique	Feb 19	Interaction	
8	Feb 24	Altair interaction	Feb 26	Interaction, contined	
		-- winter break --		-- winter break --	
9	Mar 10	Panel	Mar 12	Multidimensional	Interactive redesign (March 15th)
10	Mar 17	Design workshop	Mar 19	Geographic	Final project proposal (March 22nd)
11	Mar 24	Geographic	Mar 26	Hierarchical	
12	Mar 31	p5.js	Apr 2	Network	Final project sketches (April 5th)
13	Apr 7	p5.js continued	Apr 9	Additional topics	

14	Apr 14	Additional tools	Apr 16	Final exam	
15	Apr 21	Demo Day			Final project due (Apr 26th)

8. Course Infrastructure

Canvas (<https://umich.instructure.com>): Here you can find course materials, and submit your assignments. You will find slides and readings under "Files", and assignments under "Assignments"

Email: I will share updates and information via Announcements on Canvas. This should be sent to you UM email, so please watch for those.

Piazza: There will be a Piazza page for this course where you can ask questions or otherwise communicate with us and your colleagues.

Perusall: We will be using this for social annotations on required readings (linked from assignments page)

Sli.do (<https://www.sli.do/>): Used for collecting in-class feedback and survey responses

9. Required Readings

Most weeks will have a set of required and optional readings. The list of readings can be found in the course schedule. The readings themselves will also be available through the Canvas site.

Each week's required readings will be available on Perusall (as a single pdf). You are free to read them wherever you like, but you will need to use Perusall to provide social annotations, which are part of your grade.

Please be sure to complete the annotations by end of day on Wednesdays.

10. Visualization Tools

We will make use of a variety of visualization tools for this course, including matplotlib, Altair, p5.js, and others.

To install Altair (and python itself), we recommend using uv (<https://docs.astral.sh/uv/>), and we will cover this during one of the labs, to make sure everyone is set up.

You will be expected to use these systems for the corresponding assignments (based on weekly lab sessions), but you are free to use other software (or even sketching by hand, in some cases) for projects, design jams, critiques, and other assignments.

Other systems that you may want to explore include:

- R (<https://www.r-project.org/>) -- R is an open source, full-fledged statistical analysis environment, and has many packages available for information visualization, such as ggplot2 (<https://ggplot2.tidyverse.org>)

- Bokeh (<https://docs.bokeh.org/en/latest/>) -- Bokeh is another python package which can be useful for creating interactive visualizations
- Plotly (<https://plotly.com/>) -- Plotly is another python packages which can be used for creating interactive visualizations
- Vega and Vega-Lite (<https://vega.github.io/>) -- Vega is a declarative format for creating visualizations. Vega-lite is an even more high-level grammar for visualizations. We will not be covering these in class, but they are worth exploring if you are interested

11. Additional Recommended Resources

There are many excellent resources out there for those who would like to learn more. Here a few additional resources that I recommend:

Books:

- *Visualization Analysis and Design* by Tamara Munzner
- *The Visual Display of Quantitative Information* by Edward R. Tufte
- *How Charts Lie* by Alberto Cairo
- *Fundamentals of Data Visualization* by Clause O. Wilke
- *Data Feminism* by Catherine D'Ignazio and Lauren Klein

Blogs and Websites:

- <https://nightingaledvs.com/>
- <https://flowingdata.com/>
- <https://pudding.cool/>
- <https://www.reddit.com/r/dataisugly>

Podcasts:

- Data Stories
- The Policy Viz Podcast

12. Illness Policy

If you are feeling sick, please do not come to class. If you are concerned about aspects of these policies or about missing deadlines due to illness, let us know, and we will work with you to figure things out. Please continue to take care of yourselves and be considerate of others. Four or fewer absences will be excused automatically for the purposes of tracking attendance.

13. Accommodations for Students with Disabilities

If you think you need an accommodation for a disability, please let us know at your earliest convenience. Some aspects of this course, the assignments, the in-class activities, and the way we teach may be modified to facilitate your participation and progress. As soon as you make me aware of your needs, we can work with the Office of Services for Students with Disabilities (SSD) to help us determine appropriate accommodations. SSD (734-763-3000; <https://ssd.umich.edu/>) typically recommends accommodations through a Verified Individualized Services and Accommodations (VISA) form. I will treat any information you provide as private

and confidential.

14. Student Wellbeing

Students may experience stressors that can impact both their academic experience and their personal wellbeing. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers many resources, some of which are listed at Resources for Student Wellbeing on the Wellbeing for U-M Students website (<https://wellbeing.studentlife.umich.edu/resources-list>). You can also search for additional resources on that website.

15. Original Work

Unless otherwise specified in an assignment, all submitted work must be your own, original work. You may discuss general approaches with others on individual assignments, but may not copy code or other work and must indicate on your turned-in assignment who you worked with. You may not provide your solutions to other students. Any excerpts from the work of others must be clearly identified as a quotation, and a proper citation provided. Any violation of the School's policy on Academic and Professional Integrity (stated in the Master's and Doctoral Student Handbooks) will result in severe penalties, which might range from failing an assignment, to failing a course, to being expelled from the program, at the discretion of the instructor and the Associate Dean for Academic Affairs.

16. Policy on Generative AI

Generative AI and related tools have great potential to facilitate learning and assist in carrying out various tasks. Used in the right way, they can enhance our work, but using them improperly can border on plagiarism.

In this course, use of these tools is generally allowed, with certain exceptions. In general, we encourage you to make use of these tools for the purpose of **learning**, **brainstorming**, and **experimentation**. Remember, however, that the information they provide may not be reliable.

For weekly homework assignments: The weekly assignments are intended to challenge you and help you learn to use various tools. We will not attempt to monitor the use of AI for completing these assignments, but strongly encourage you to attempt to figure out the assignments on your own, and use them as an opportunity to learn.

For projects: In creating visualizations for the projects, you are allowed to make use of AI tools as assistants in creating visualizations. However, it is important to remember that these tools can be useful assistants, but should not be used as ghostwriters. You are ultimately accountable for the material in the project reports.

For social annotation of weekly readings: This is the one case where you are prohibited from using generative AI. We want to know what you think about these readings, not what an LLM thinks. There are no

wrong answers here, but it is important to engage with the material and share your thoughts.

17. Frequently Asked Questions

What if I am waitlisted?

You are welcome to attend lectures and labs while waiting for a space to open up. The add/drop deadline is January 27th.

Do I have to attend the lectures? What about the labs?

Yes, regular attendance is expected for both. If you will be unable to attend the lecture or the lab most weeks, you should probably reconsider taking this section of this course this term. Up to four absences will be excused automatically. However, you will need to be in class in person on April 16th in order to take the final exam. You are strongly advised to arrange your schedule so as to be in Ann Arbor on that day.

What about Tableau?

For various reasons we'll discuss in class, we'll be emphasizing other tools this term, particularly programmatic tools that use python or javascript to convert data into visualizations. We may touch briefly on Tableau towards the end of the course, but it will not be a priority.

What about D3?

D3 is often thought of as a package for data visualization, but it is really more of a collection of packages, most of which deal with data *manipulation*. D3 is very powerful, but also unfortunately has a steep learning curve. We may touch on it towards the end of the term, but it will also not be a priority.